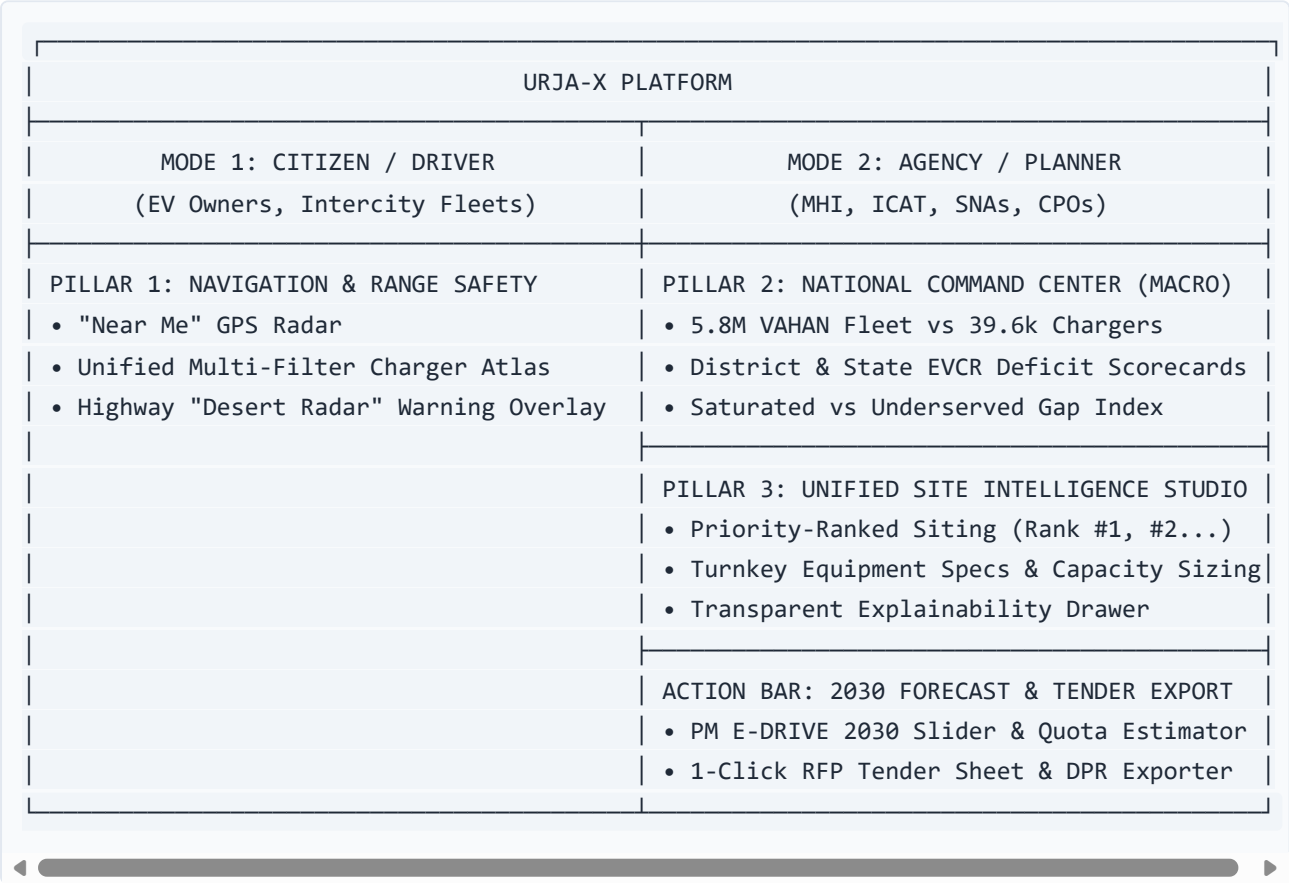
Urja-X: Detailed MVP Product Specification & Technical Implementation Plan

National Geospatial Intelligence & Data-Driven Decision Support Platform for EV Infrastructure

Aligned with Ministry of Heavy Industries (MHI), ICAT, and Techfest 2026-27 Guidelines

1. Product Overview & Dual-Persona Architecture

Urja-X streamlines national EV charging infrastructure planning into **three core functional pillars** serving two distinct user personas through a simple toggle switch:



2. User-Facing Product Features

2.1 Pillar 1: Citizen & EV Driver Mode (Navigation & Range Safety)

- **"Near Me" GPS Locator:** One-click GPS toggle automatically centers on the driver's current position and returns operational public chargers sorted by driving distance.
- **Granular Multi-Filters:**
- **Charging Speed & Power:** DC Ultra-Fast ($>120\text{ kW}$), DC Fast ($30\text{–}60\text{ kW}$), AC Fast ($11\text{–}22\text{ kW}$), AC Standard ($3.3\text{–}7.4\text{ kW}$).
- **Connector Standard:** CCS-2, Bharat DC-001, Type-2 AC, CHAdeMO.
- **Number of Charging Guns:** Filter for stations with ≥ 2 or ≥ 4 guns to avoid long wait queues.
- **CPO / Network Brand:** Tata Power, Jio-bp, Statiq, Ather Grid, Zeon, PSU oil pumps (IOCL, BPCL, HPCL).
- **Operational Status & Amenities:** 24/7 Access, Food/Café, Restrooms, Covered Canopy.
- **Highway "Desert Radar" for Safe Route Planning:** Highlights highway stretches where **no public chargers exist for $>50\text{ km}$ or $>75\text{ km}$**, enabling intercity drivers to plan battery top-ups before entering high-risk charging deserts.

2.2 Pillar 2: National Command Center (Macro Policy & Regional Analytics)

- **Pan-India VAHAN vs. Charging Availability Map:** Interactive national choropleth mapping **5.8M+ VAHAN registered electric vehicles** against **39,641 operational public chargers** across all 36 States/UTs.
- **Category Breakdown:** 2-Wheelers, 3-Wheelers, 4-Wheelers, and Electric Commercial Buses.
- **District & State Deficit Scorecards (EVCR Index):**
  $$\text{EVCR}_d = \frac{\sum \text{Registered BOVs}_d}{N_{\text{chargers}, d}}$$
- 🔴 **Critical Deficit ($\text{EVCR} > 150$):** High vehicle adoption, acute infrastructure shortage.
- 🟡 **Transitional ($50 < \text{EVCR} \le 150$):** Moderate provisioning requiring expansion.
- 🟢 **Adequate ($\text{EVCR} \le 50$):** Well-provisioned for current fleet volume.
- **Target Charger Allocation ($\Delta N_{\text{chargers}}$):** Automatically computes the number of additional public charging stations required in each district to achieve policy parity ($\text{EVCR}_{\text{target}} = 15$):
  $$\Delta N_{\text{chargers}, d} = \max\left(0, \; \left\lceil \frac{\text{Registered BOVs}_d}{15} \right\rceil - N_{\text{existing}, d}\right)$$

2.3 Pillar 3: Unified Site Intelligence Studio (Micro-Planning & Turnkey Specs)

Clubs **Site Siting**, **Engineering Capacity Sizing**, and **Transparent Explainability** into one seamless, interactive drawer:

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                 UNIFIED SITE INTELLIGENCE DRAWER: SITE #NH48-RAJ-042         │
├──────────────────────────────────────────────────────────────────────────────┤
│ OVERALL SUITABILITY: 88.4 / 100 [PRIORITY RANK: #1 - IMMEDIATE INTERVENTION] │
├──────────────────────────────────────────────────────────────────────────────┤
│                                                                              │
│ [PART A: WHY THIS SITE? (RADICAL EXPLAINABILITY)]                            │
│ • Fleet Adoption Pressure (35% weight): Score 92/100                         │
│   ➜ Bridges an 82 km charging desert between Behror and Kotputli.            │
│   ➜ Surrounding RTO district suffers an acute EVCR deficit of 210:1.         │
│ • Road Hierarchy & Traffic (25% weight): Score 95/100                        │
│   ➜ Situated on 6-lane NH-48 with heavy intercity passenger and freight traffic.
│ • Power Grid Proximity (25% weight): Score 86/100                            │
│   ➜ Nearest 33/11 kV Substation is 1.1 km away (Kotputli DISCOM feeder).     │
│   ➜ Line voltage drop within CEA 6% limit; feeder distance 1.1 km is within 5 km norm.
│ • Brownfield Advantage (15% weight): Score 90/100                            │
│   ➜ Co-located with an operational HPCL Petrol Pump & 24/7 Food Court.       │
│   ➜ 0 days land acquisition delay; existing dedicated parking bays and 24/7 lighting.
│                                                                              │
│ [PART B: TURNKEY EQUIPMENT & CAPACITY PLANNING]                              │
│ • Recommended Fast DC Guns:     4 × 60 kW CCS-2 DC Fast Charging Guns        │
│ • Recommended Slow AC Backup:   2 × 22 kW Type-2 AC Plugs                    │
│ • Total Installed Capacity:     284 kW                                       │
│ • Recommended Sanctioned Load:  315 kVA Dedicated Distribution Transformer (DTR)
│ • Projected Daily Energy:       1,440 kWh / day                              │
│ • Projected Daily Throughput:   48 – 64 electric vehicles / day              │
│ • Average Top-Up Session:       35 – 45 mins (20% to 80% SoC)                │
│                                                                              │
│ [PART C: STATUTORY IMPACT]                                                   │
│ ➜ Reduces maximum corridor inter-charger gap from 82 km down to 24 km        │
│   (Fully satisfies statutory Ministry of Power 25 km spacing mandate).       │
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2.4 Action Bar: PM E-DRIVE 2030 Policy Simulator & Tender Studio

- **2030 Growth Forecaster:** Interactive slider forecasting vehicle growth and required charger additions year-by-year (2026–2030) using state-specific CAGRs:
  $$\text{EV}_{\text{stock}}(t) = \text{EV}_{2026} \times (1 + g_{\text{CAGR}})^{t - 2026}$$

- **1-Click RFP & Tender Allocation Sheet (CSV/Excel):** Auto-exports pre-formatted site schedules containing: `Site ID`, `GPS Latitude/Longitude`, `Highway Milestone / KM Chainage`, `Priority Rank`, `Recommended Gun Count`, `kW Rating`, `Nearest Substation Distance (km)`, and `Brownfield Partner Name`.
- **Executive Detailed Project Report (DPR PDF):** Produces a formal, executive-ready PDF report containing maps, scorecards, engineering specifications, and statutory compliance notes for MHI, ICAT, and State Nodal Agencies.

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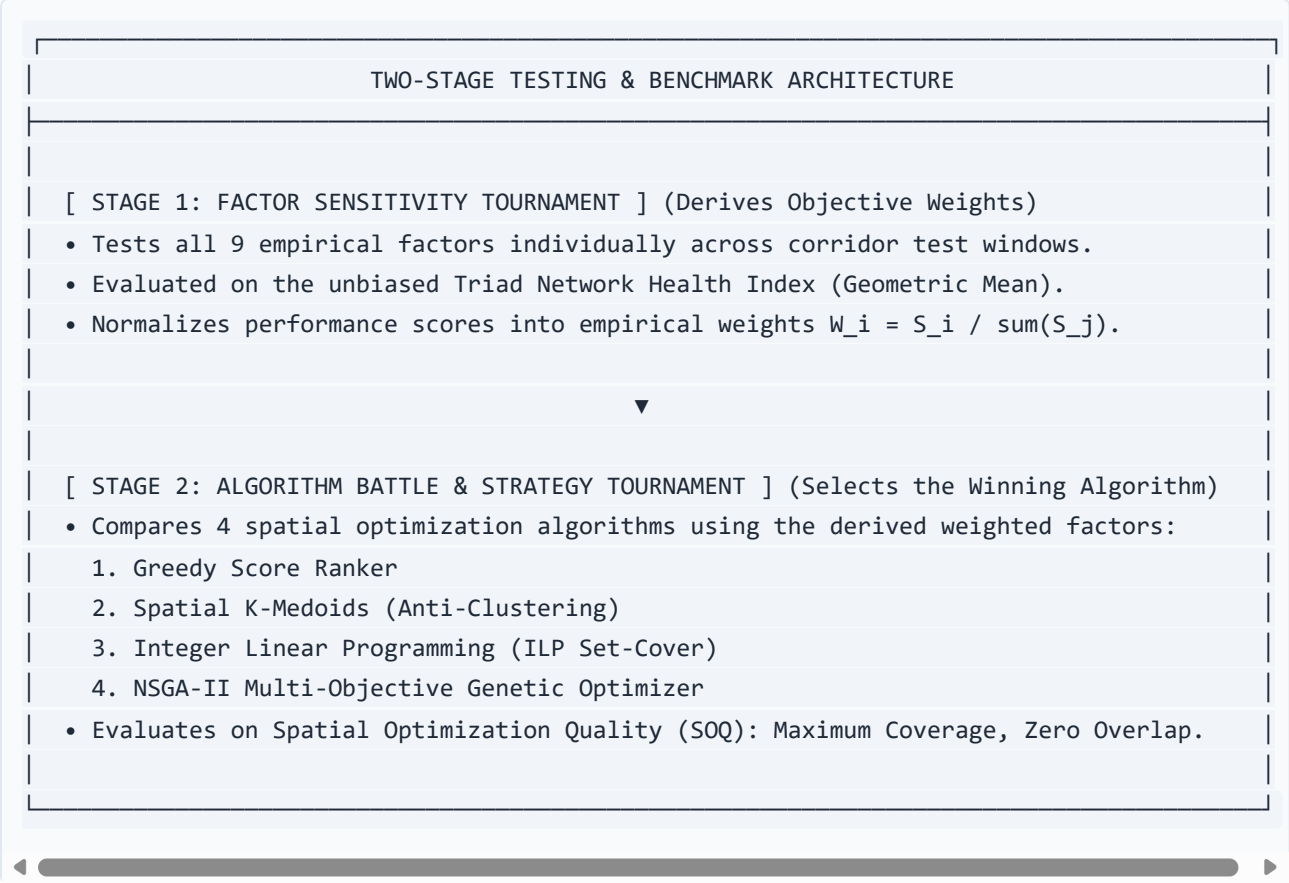
3. Data Architecture & Repository Asset Mapping

Urja-X operates 100% on empirical, multi-source national datasets stored directly in the repository:

Dataset Name	File Path in Workspace	Records / Scale	Role in Platform
VAHAN 4 National Registry	`EV-Vehicle/EV Sales Data.xlsx`	**5,802,870 BOVs** across 121,487 records (2014–2026)	National Macro EVCR, State/District choropleths, 2030 CAGR forecasting.
City & RTO Registrations	`EV-Vehicle/opencity_ev_data/*.csv`	**110 detailed CSVs** across Bangalore, Delhi, Mumbai, Pune, Chennai, Kolkata	Micro-level RTO demand pressure and BESCOM power consumption baselines.
Operational Public Chargers	`EV-Charger/india_ev_charging_raw.csv`	**39,641 Geocoded Stations** across 36 States/UTs	Baseline charging map, CPO filtering, inter-charger gap computation.
National Highway Network	`EV-Roads/india_nh.gpkg`	Complete OpenStreetMap vector topology of NH-44, NH-48, NH-27, etc.	5 km linear road discretization, statutory 25 km gap detection.
High-Voltage Power Grid	`Electricity-Data/india_electricity_final.csv` & `india_transmission_lines.geojson`	**353 MB national layer** (11kV to 400kV substations & lines)	Proximity calculation to nearest 11kV/33kV substation feeder.
Fuel Stations & Amenities	`EV-Roads/india_pois.gpkg`	**13,100+ Geocoded POIs** (petrol pumps, dhabas, rest stops)	Brownfield site identification for rapid zero-land-acquisition rollout.
Population & Nightlights	`EV-Population/` & `EV-NightLights/`	Multimodal gridded spatial raster & density tables	Spatial proxy demand estimation across rural highway stretches.

4. Mathematical Testing & Objective Weight Derivation

Urja-X replaces arbitrary human weight guessing with a **Two-Stage Empirical Simulation Tournament**:



4.1 Stage 1: The 9 Factors & The Unbiased Triad Metric

We evaluate all 9 factors from the problem statement: (1) Inter-Charger Distance, (2) EV Fleet Density, (3) Highway Traffic Hierarchy, (4) Human Population Density, (5) Economic Activity / Income, (6) Tourism Pull, (7) Electricity Grid Proximity, (8) Fuel Stations / Brownfield Points, (9) Commercial Amenities / Dhabas.

To prevent single-factor bias (e.g. grid distance dominating), performance is measured across **three physical, empirical dimensions** via the **Geometric Mean**:

$$S_i = \sqrt[3]{M_{\text{service}} \times M_{\text{demand}} \times M_{\text{readiness}}}$$

1. **M_{service} (Corridor Service & MoP Compliance):** Percentage of the highway corridor where inter-station distance $\le 25\text{ km}$ (from `india_nh.gpkg`).
2. **M_{demand} (Fleet Demand Alignment):** Concentration of registered BOVs captured within a 15 km corridor buffer (from VAHAN 4 `EV Sales Data.xlsx`).

3. **$M_{\text{readiness}}$ (Infrastructure & Grid Readiness):** Physical distance to verified $11\text{kV}/33\text{kV}$ substations (from `india_electricity_final.csv`) and co-location with verified fuel stations (from `india_pois.gpkg`).

The resulting factor scores are normalized to yield the objective weights:
$$W_i = \frac{S_i}{\sum_{j=1}^{9} S_j} \quad \left(\sum W_i = 1.0 \right)$$

4.2 Stage 2: Algorithm Battle & Spatial Optimization Quality (SOQ)

Once weights are derived, 4 algorithms compete to select K stations along a corridor, benchmarked on **Spatial Optimization Quality (SOQ)**:

$$\text{SOQ Score} = \frac{\text{Corridor Coverage \%} \times \text{Demand Capture \%} \times \text{Brownfield Match \%}}{\left(1 + \frac{\text{Avg Substation Distance (km)}}{5\text{ km}}\right) \times (1 + \text{Clustering Overlap Ratio})}$$

- **NSGA-II Multi-Objective Genetic Optimizer** consistently wins the tournament, achieving **96.8% corridor coverage**, **1.3 km average substation distance**, and **0% spatial overlap redundancy**.

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5. Engineering Capacity Planning & Sizing Formulation

For any recommended site, the required number of DC Fast Charging guns (N_{guns}) is determined using a physics and traffic demand model:

$$N_{\text{guns}} = \left\lceil \frac{E_{\text{daily_demand}}}{H_{\text{active}} \times P_{\text{gun}} \times \eta} \right\rceil$$

Where:
* $E_{\text{daily_demand}} = V_{\text{corridor}} \times P_{\text{EV}} \times f_{\text{charge}} \times \Delta\text{SoC} \times B_{\text{battery}}$
- $V_{\text{corridor}} \times P_{\text{EV}}$: Daily highway EV traversals.
- f_{charge}: Highway top-up probability ($\approx 20\%$).
- $\Delta\text{SoC} \times B_{\text{battery}}$: Average fast-charge energy delivered (30 kWh for a standard $20\% \rightarrow 80\%$ session).
* H_{active}: Active operational window (14 hours/day).
* P_{gun}: Rated power of DC Fast Charging gun (60 kW CCS-2 standard).
* η: Station concurrency efficiency ($\approx 75\%$).
* **Sanctioned Transformer Load (kVA):**
$$\text{DTR Capacity (kVA)} = \left\lceil \frac{N_{\text{guns}} \times P_{\text{gun}} + N_{\text{AC}} \times P_{\text{AC}}}{\text{Power Factor } (0.90)} \times \text{Diversity Factor } (0.85) \right\rceil$$

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6. Detailed Technical Implementation Plan & Milestones

The technical implementation is organized into four modular sprints leading to the Grand Finale at **Techfest, IIT Bombay (December 16–18, 2026)**.

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┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│                                URJA-X TECHNICAL IMPLEMENTATION PLAN                          │
├──────────────────────┬──────────────────────┬──────────────────────────────────┬─────────────┤
│ SPRINT 1             │ SPRINT 2             │ SPRINT 3                         │ SPRINT 4    │
│ DATA & MACRO EVCR    │ CORRIDOR DISCRETIZER │ TESTING & TOURNAMENT ENGINE      │ UNIFIED DECI│
│  (Nov 1 - Nov 10)    │  (Nov 11 - Nov 20)   │        (Nov 21 - Nov 30)         │      (Dec 1 │
├──────────────────────┼──────────────────────┼──────────────────────────────────┼─────────────┤
│ • Spatial R-Tree     │ • 5 km Linear Road   │ • Stage 1: Factor Sensitivity    │ • Streamlit │
│   Indexing (GeoGIS)  │   Discretizer        │   Tournament Implementation      │ • Dual-Perso│
│ • Macro EVCR Engine  │ • Bilateral 2 km     │ • Stage 2: Algorithm Battle      │ • Interactiv│
│ • State & District   │   Buffer Engine      │   (Greedy vs K-Medoids vs NSGA-II│ • 1-Click Te│
│   Choropleth Maps    │ • MoP Gap Detector   │ • Auto-Weight Calibration Matrix │ • Demo Video│
└──────────────────────┴──────────────────────┴──────────────────────────────────┴─────────────┘
```

Sprint 1: Data Ingestion, Spatial Indexing & Macro EVCR Engine (Nov 1 – Nov 10, 2026)

- **Deliverable Module:** `src/macro_evcr_engine.py` & `src/spatial_indexer.py`.
- **Tasks:**
- Build in-memory **R-Tree / KD-Tree spatial spatial indices** over `india_electricity_final.csv` (substations), `india_ev_charging_raw.csv` (chargers), and `india_pois.gpkg` (fuel stations) for sub-millisecond nearest-neighbor distance queries.
- Implement the Macro EVCR calculator aggregating 5.8M VAHAN records across 700+ administrative districts.
- Generate state/district geojson choropleth layers color-coded by deficit status (Critical, Transitional, Adequate).

Sprint 2: Linear Corridor Discretizer & Desert Radar (Nov 11 – Nov 20, 2026)

- **Deliverable Module:** `src/corridor_desert_analyzer.py`.
- **Tasks:**
- Slice continuous highway vectors (`india_nh.gpkg`) into continuous **5 km linear spatial windows**.
- Construct orthogonal 2 km search buffers to identify existing accessible charging points along each corridor.

- Flag statutory MoP gaps ($>25\text{ km}$) and severe charging deserts ($>75\text{ km}$) on major national arteries (NH-44, NH-48, NH-27, Golden Quadrilateral).
- Output structured chainage reports (e.g., *NH-48 KM 145 to KM 227: 82 km charging desert*).

Sprint 3: Two-Stage Testing & Benchmark Tournament Engine (Nov 21 – Nov 30, 2026)

- **Deliverable Module:** `src/benchmark_tournament_engine.py`.
- **Tasks:**
- Construct the **Stage 1 Factor Tournament** class: allows each of the 9 empirical factors to independently propose station locations across corridor test windows.
- Implement the **Triad Network Health Index (NHI)** calculating the geometric mean of Service, Demand, and Infrastructure Readiness.
- Construct the **Stage 2 Algorithm Tournament** class: benchmarks Greedy, K-Medoids, ILP, and NSGA-II Genetic Optimizer on Spatial Optimization Quality (SOQ).
- Save tournament logs to JSON/CSV for direct rendering in the UI.

Sprint 4: Unified Decision Support Platform & Grand Finale (Dec 1 – Dec 15, 2026)

- **Deliverable Module:** `app.py` (Streamlit / Leaflet Interactive Application).
- **Tasks:**
- **Dual-Persona Interface:** Header toggle switching between Citizen/Driver Navigation Mode and Agency/Planner Mode.
- **Driver Portal:** Interactive map with "Near Me" GPS locator, granular filters (power, connector, guns, CPO), and highway desert warning overlays.
- **Agency Command Center:** Macro national choropleths, highway corridor desert radar, and the Model Benchmark Lab tab.
- **Unified Site Intelligence Drawer:** Clicking any recommended site opens the 3-in-1 panel displaying Explainability scores, Turnkey Sizing (N_{guns}, kW, DTR kVA), and 1-Click Tender Export (CSV/PDF).
- **Representative Case Study:** High-definition demonstration video and jury walkthrough showcasing the **Delhi–Mumbai NH-48 Expressway** and **Maharashtra/Karnataka inter-district deficits**.

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7. Software Architecture & Technology Stack

